21 · Rapid-fire: every case answer, back to back

The words of this track, as it says them. Bold brown is the examiner.

This is the rapid-fire track: every case answer in this episode, back to back, one breath each, in booklet order. It is made for the morning of the exam. If one of these sounds unfamiliar, go back to its own track.

Case 2, lymph node removal.

The lymphatic system has three jobs. Digestive: lacteals absorb dietary fat. Immunological: the thymus, the bone marrow and the lymph nodes. And fluid balance: draining excess tissue fluid back to the venous system. Cancer spreading is metastasis, and cells in the lymph can reach any part of the body. The axillary nodes are checked, the sentinel node first: cancer there means it has spread; none means it's likely limited to the breast. Removing the nodes blocks the arm's drainage to the subclavian vein, so fluid stays in the tissue: lymphoedema.

Case 3, the asthma attack.

Difficulty breathing is dyspnoea: dys, bad, pnoea, breathing. Three things narrow Ane's airways: bronchoconstriction, with the smooth muscle contracted; inflammation, a swollen airway wall; and more mucus. Airway resistance is high, so it needs a greater pressure difference, and the accessory muscles are recruited: more work of breathing. The reliever is a bronchodilator: it relaxes the airway smooth muscle so the airway widens. A preventer is a corticosteroid, an anti-inflammatory.

Case 3, the airway defences and blood pressure.

The defences: mucus and cilia, the muco-ciliary escalator; nasal hairs that filter; the carina, which triggers the cough reflex; and the epiglottis, for large objects. The escalator: cilia on the respiratory epithelium beat to sweep sticky, dust-trapping mucus up and out. Smoking damages the cilia, so mucus can't be moved, and a smoker's cough replaces them. Blood pressure: the carotid and aortic baroreceptors sense the stretch and tell the medulla oblongata. The cardio-inhibitory centre is stimulated and the cardio-acceleratory centre inhibited, and the vagus nerve slows the rate and force, so cardiac output drops. The vasomotor centre causes vasodilation, and the pressure returns to normal.

Case 3, the four lung micrographs.

The trachea and a bronchus: ciliated pseudostratified columnar epithelium, with cartilage and smooth muscle. A bronchiole: cuboidal epithelium, with smooth muscle but no cartilage. The alveoli: type one pneumocytes, squamous epithelium for gas exchange; type two pneumocytes for surfactant; macrophages to keep them clean; and no cartilage or muscle. Bronchodilators relax the smooth muscle; daily cortisol, the preventer, reduces the inflammation.

Case 4, breathing in and out.

Inhalation: the diaphragm and external intercostals contract, the thoracic cavity enlarges, the intrapleural and then intrapulmonary pressures fall, by Boyle's law, and air flows in from high pressure to low. Exhalation: they relax, the elastic lungs recoil passively, intrapulmonary pressure rises, and air flows out. Forced breathing in adds the sternocleidomastoid and scalenes; forced breathing out uses the abdominal muscles and internal intercostals. Ventilation gets harder with blocked airways, with stiff lungs from scar tissue or no surfactant, and with a lost pleural seal: a pneumothorax collapses the lung.

Case 4, breathing and blood pH.

Normal blood pH is seven point three five to seven point four five. The buffer: carbon dioxide plus water, carbonic acid, hydrogen ions plus bicarbonate, both ways. Breathing sets the blood's carbon dioxide, and hydrogen ions follow it, so pH follows too. Manaia's hyperventilation blows off carbon dioxide, hydrogen ions fall, and pH rises: respiratory alkalosis. Hypoventilation is the reverse: respiratory acidosis. In metabolic acidosis the body hyperventilates, to breathe off the extra carbon dioxide; in metabolic alkalosis it hypoventilates, to keep it.

Case 4, the two acid-base tables.

Metabolic acidosis: hydrogen ions up; the buffer converts them into carbon dioxide and water, so carbon dioxide rises, and breathing speeds up. Metabolic alkalosis: hydrogen ions down; the buffer makes them from carbon dioxide and water, so carbon dioxide falls, and breathing slows. Hyperventilation: carbon dioxide down; the buffer uses up hydrogen ions, so they fall and pH rises: alkalosis. Hypoventilation: carbon dioxide up; the buffer makes hydrogen ions, so pH falls: acidosis. Compare them: it's the same buffer, and carbon dioxide and hydrogen ions always move together.

Case 6, what insulin does.

Insulin is made by the beta cells of the islets of Langerhans, in the endocrine pancreas, and it lowers blood sugar; the alpha cells make glucagon. Its opponents: glucagon, adrenaline from the adrenal medulla, cortisol from the adrenal cortex, growth hormone, and the thyroid hormones. At a cell, insulin travels in the blood, binds its receptor, and a cascade moves GLUT 4 into the membrane, letting glucose in by facilitated diffusion. In the liver and muscles it drives glycogenesis, glucose to glycogen. In fat tissue, lipogenesis.

Case 6, glucagon and the day graph.

Glucagon raises blood glucose three ways: glycogenolysis, lipolysis, and gluconeogenesis in the liver. Over the day, blood glucose follows your meals, but it never hits zero and stays in the normal range, and insulin never hits zero either. Basal insulin is the background; bolus insulin comes after meals. Normally, blood glucose never goes above the normal range. It's lowest overnight, so breakfast breaks the fast. Insulin resistance overworks the pancreas until it fatigues and makes too little insulin: type 2 diabetes, and possibly insulin injections.

Case 8, the joint.

The joint, letter by letter: muscle contracts to move the bones; the enthesis is where ligaments attach; synovial fluid lubricates; a bursa reduces friction; the joint capsule holds the bones together; a tendon joins muscle to bone; bone is a lever that articulates; hyaline cartilage reduces friction and absorbs shock; and a ligament holds bones together. Synovial joints are freely movable. The shoulder and hip are ball-and-socket; the knee and elbow are hinges, like a door. Bursitis is an inflamed, swollen bursa, the friction cushion, and it's painful.

Case 8, rheumatoid arthritis and cortisol.

Rheumatoid arthritis is autoimmune. It comes on over weeks to months, at any age, affects joints on both sides with morning stiffness over an hour, and brings whole-body fatigue. Osteoarthritis is wear and tear: later in life, over years, one set of joints, little swelling, stiffness under an hour, and no whole-body symptoms. Cortisol comes from the adrenal cortex, released by ACTH. It raises blood glucose by gluconeogenesis, keeps blood volume and perfusion up through long-term stress, reduces inflammation, and suppresses the immune system, which is how it helps Shannon. And tell her: it's autoimmune, and it can hit at any age.

Case 9, the synapse.

A neurotransmitter is a chemical released at the presynaptic endings of a chemical synapse, which acts on the receptors of the postsynaptic neuron. Dopamine is made in the substantia nigra of the midbrain. The action potential reaches the axon terminal, calcium channels open, and calcium makes the vesicles release neurotransmitter into the cleft by exocytosis. It diffuses across, binds receptors on the postsynaptic neuron, and changes the membrane's permeability: excitatory, or inhibitory. Then it's cleared, by re-uptake or enzymatic destruction.

Case 9, the synapse labels and levodopa.

Letter A, a calcium channel. B, a vesicle of neurotransmitter. C, the neurotransmitter molecules. D, a reuptake channel. E, the receptors. F, the presynaptic neuron, with the vesicles. G, the postsynaptic neuron, with the receptors. Into the cleft by exocytosis, and across it by diffusion. Levodopa, not dopamine, because a drug must cross the blood-brain barrier to work on the brain. Levodopa crosses and is made into dopamine there; dopamine itself can't cross.

Case 10, stroke and the brain.

Area A, the prefrontal cortex: judgement, so damage brings poor judgement and inappropriate behaviour. B, Broca's area: producing speech. C, the motor area, on the precentral gyrus: voluntary movement. D, the somatosensory area, on the postcentral gyrus: touch. E, the auditory area: hearing. F, Wernicke's area: understanding language. G, the occipital area: vision. Lose the area, lose its job. Drunk with no alcohol: the cerebellum. Long, gasping in-breaths: the pons. Very slow, shallow breathing: the medulla oblongata. Hands acting independently: the corpus callosum.

Case 13, ovulation and the cycle graph.

The ovulation predictor kit catches the peak in LH in the urine, just before ovulation. At ovulation, an LH peak makes the Graafian follicle burst, the ovarian wall ruptures, and a secondary oocyte is pushed out toward the fimbriae of the fallopian tube. The fertile days: ten to seventeen. Each hormone drives its own process: FSH grows the follicles, oestrogen thickens the lining, LH triggers ovulation, and progesterone prepares for implantation. In pregnancy, the drop at about day twenty-one doesn't happen: oestrogen and progesterone keep rising.

Case 13, contraception and the open questions.

Progesterone contraception stops ovulation by blocking LH, thickens the cervical mucus to keep sperm out, and imitates pregnancy. Hormonal methods, the pill, the Jadelle implant and the mini pill, prevent pregnancy only. Condoms are a barrier, and the only method that also stops STIs, though not pubic lice. The IUD may prevent implantation, and prevents pregnancy only. Fertility needs healthy sperm and hormones in a male, and regular ovulation, open tubes and a healthy lining in a female. IVF fertilises eggs in a lab and places the embryo in the uterus, and the FSH injections grow several follicles at once, so more eggs can be collected.

Case 14, rods, cones and colour blindness.

Cones: colour, central vision, in good light, with more detail: photopic. Rods: the grey scale, more sensitive to light, working in dim light and for peripheral vision: scotopic. The macula has both; the fovea has only cones. Red-green colour blindness is a genetic defect of the retina's cones: faulty L cones, for long wavelength red light, and M cones, for medium wavelength green, so red and green are hard to tell apart. A colour-blindness test detects it.

Case 14, Simon's family.

Grandfather, X little b, Y: colour blind. Grandmother, X big B, X little b: a carrier, because she has a colour-blind daughter. Aunt, X little b, X little b. Uncle, X big B, Y. Mum, X big B, X little b: a carrier, with normal vision. Dad, X big B, Y. Simon, X little b, Y: his X came from his mum, and his Y from his dad. It's X-linked recessive: carried by mothers, shown in half of their sons, a woman needs both X's, fathers never pass it to sons and always pass it to daughters, and it can skip a generation. Simon will pass it to none of his sons, and to all of his daughters, as unaffected carriers.

Case 15, genetic diseases.

Cystic fibrosis is autosomal recessive, on chromosome seven: it skips generations, there are carriers, and the affected person is little r little r. Huntington's is autosomal dominant, on chromosome four: it shows in every generation, and the affected person is big R little r, or big R big R. On a recessive chart, label the affected people little r little r first; their unaffected parents are carriers. A child is affected only if both parents pass on a little r: two carriers give twenty-five per cent. On a dominant chart, affected is big H little h and unaffected is little h little h. An unaffected cousin can't pass it on; an affected one has a fifty per cent chance with

each child.

That's every answer. Go and get them.